SERBAN EUGEN CICORTAS

Courant Institute of Mathematical Sciences, 251 Mercer Street, New York, NY 10012
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EMPLOYMENT

Courant Institute, New York University **August 2026 - Present**
Courant Instructor

EDUCATION

Princeton University **September 2021 - May 2026**
PhD in Mathematics
PhD Thesis: Nonlinear Scattering Theory for Asymptotically de Sitter Vacuum Solutions in All Even Spatial Dimensions.
Advisor: Igor Rodnianski.

University of Cambridge, Trinity College **October 2020 - July 2021**
Master of Advanced Study in Pure Mathematics, Distinction
Part III Essay: The Existence of Naked Singularities and Their Instability.
Advisor: Mihalis Dafermos.

Princeton University **September 2016 - June 2020**
Bachelor of Arts in Mathematics, Highest Honors
Senior Thesis: Global Stability for Weak Null Wave Equations with Multi-localized Initial Data.
Advisor: Sergiu Klainerman.

RESEARCH INTERESTS

Partial Differential Equations, General Relativity, Nonlinear Wave Equations, Compressible Fluids

PUBLICATIONS

7. Instability of self-similar implosions for the compressible Euler equations 2026

with Igor Rodnianski and Ely Sandine
arXiv:2611.07762.

6. Weak cosmic censorship for the circularly symmetric Einstein-scalar field system in 2+1 dimensions 2026

arXiv:2605.19143.

5. Discretely self-similar exterior-naked singularities for the Einstein-scalar field system 2024

with Christoph Kehle
Arch Rational Mech Anal 250, 37 (2026). arXiv:2412.09540.

4. Systems of wave equations on asymptotically de Sitter vacuum spacetimes in all even spatial dimensions 2024

Journal of Functional Analysis 290, no. 11 (2026). arXiv:2411.16655.

3. Nonlinear scattering theory for asymptotically de Sitter vacuum solutions in all even spatial dimensions 2024

arXiv:2410.01558. To appear in *Journal of Differential Geometry*.

2. Scattering for the wave equation on de Sitter space in all even spatial dimensions 2023

arXiv:2309.07342.

1. Extensions of Lorentzian Hawking–Page solutions with null singularities, spacelike singularities, and Cauchy horizons of Taub–NUT type 2022

Ann. Henri Poincaré 26, 3907–3961 (2025). arXiv:2209.06134.

INVITED TALKS

Analysis Seminar, Princeton University, Upcoming talk	November 2026
Analysis Seminar, Stony Brook University, Upcoming talk	November 2026
Analysis Seminar, NYU	October 2026
Princeton Gravity Initiative seminar, Princeton University	September 2026
"Singularities and Cosmic Censorship, from vacuum to matter" Workshop, Fields Institute	May 2026
Hyperbolic and Dispersive PDE seminar, Rutgers University	March 2026
Analysis Seminar, Yale University	February 2026
Geometric Analysis and PDE seminar, University of Cambridge	February 2026
Junior Analysis Seminar, Imperial College London	February 2026
PDE Seminar, Pennsylvania State University	January 2026
ICERM Extremal Black Holes and the Third Law of Black Hole Thermodynamics	January 2026
Analysis and PDE Seminar, UC Berkeley	November 2025
Analysis and PDE Seminar, Stanford University	November 2025
Analysis Seminar, Columbia University	October 2025
PDE/Analysis Seminar, MIT	September 2025
Analysis Seminar, Cornell University	September 2025
Hyperbolic and Dispersive Equations on Curved Geometries Workshop, Simons Center	April 2025
Elaine Hubbard PDE Seminar, Georgia Institute of Technology	November 2024
Analysis Seminar, University of Toronto	November 2024
Princeton Gravity Initiative Workshop, Flash talk	March 2024

Princeton Graduate Student Seminar February 2024

Mathematical GR and Hyperbolic PDE Seminar, Online talk March 2023

Princeton Graduate Student Seminar February 2023

TEACHING EXPERIENCE

Instructor for MATH-UA 325 - Analysis Fall 2026

Teaching Assistant for Integration Theory, Complex Analysis Fall 2024 - Spring 2026

Instructor for SLMath Summer School
"Mathematics of General Relativity and Fluids" Summer 2024

Preceptor for MAT 201 - Multivariable Calculus Summer 2024

Preceptor for MAT 175 - Mathematics for Economics/Life Sciences Fall 2023 - Spring 2024

Teaching Assistant for Integration Theory, Complex Analysis Fall 2022 - Spring 2023

AWARDS

Bradford H. Arnold *42 and Mary Ellen Arnold Fellowship 2025 - 2026

Centennial Fellowship 2021 - 2025

Trinity Studentship in Mathematics 2020 - 2021

George B. Covington Prize in Mathematics 2020

Elected to Membership in the Society of Sigma Xi and Phi Beta Kappa Society 2020

Peter A. Greenberg '77 Prize 2019

Princeton University German Department Book Prize 2017, 2018, 2019

SERVICE

Referee for:
Annales Henri Poincaré, Communications in Mathematical Physics, Annals of PDE

Princeton-Cambridge Joint General Relativity Seminar Organizer 2022-2025

Princeton Graduate Student Seminar Co-Organizer 2022-2023